

Ans to the Q.no-1

$$X = (9.321)_{10}$$
$$= (1001.0101)_2$$

$$Y = (3.325)_{10}$$
$$= (11.0101)_2$$

Normalizing both X and Y we get,

$$X = 1.0010101 \times 2^3$$

$$Y = 1.10101 \times 2^1$$

Now, $x * y$

$$= (1.0010101 \times 2^3) \times (1.10101 \times 2^1)$$

$$= (1.0010101 \times 1.10101) \times 2^4$$

$$= 1.111011011001 \times 2^4$$

(Ans)

$$0.321 \times 2 = 0.642$$

$$0.642 \times 2 = 1.284$$

$$0.284 \times 2 = 0.568$$

$$0.568 \times 2 = 1.136$$

$$0.325 \times 2 = 0.65$$

$$0.65 \times 2 = 1.3$$

$$0.3 \times 2 = 0.6$$

$$0.6 \times 2 = 1.2$$

Ans to the Q: no-2

$$18.7110 - (-7.0110) = 18.7110 + 7.0110$$

$$(18.7110)_{10} = (10010.1011)_2$$

$$(7.0110)_{10} = (111.0000)_2$$

Normalizing both numbers we get,

$$(10010.1011)_2 = 1.00101011 \times 2^4$$

$$(111.0000)_2 = 1.11 \times 2^2$$

$$0.7110 \times 2 = 1.422$$

$$0.422 \times 2 = 0.844$$

$$0.844 \times 2 = 1.688$$

$$0.688 \times 2 = 1.376$$

$$0.0110 \times 2 = 0.022$$

$$0.022 \times 2 = 0.044$$

$$0.044 \times 2 = 0.088$$

$$0.088 \times 2 = 0.176$$

$$\text{Now, } (1.00101011 \times 2^4) + (1.11 \times 2^2)$$

$$= (1.00101011 \times 2^4) + (0.0111 \times 2^4)$$

$$= (1.00101011 + 0.0111) \times 2^4$$

$$= 1.10011011 \times 2^4$$

(Ans)

$$\text{Bias} = 2^{8-1} - 1 = 127$$

$$\text{Biased Exponent} = 127 + 4 = 131$$

$$\text{Range of Bias exponent} = 0 \text{ to } 2^8 - 1$$

$$= 0 \text{ to } 255$$

$$= 1 \text{ to } 254$$

$$1 < 131 < 254$$

No overflow or underflow